MSET005

Question 1.

[Tg:@abusat_760] Given $f(x) = -3x^2 + 5$ for all real numbers, what is the range of the function?

A) All real numbers less than or equal to 5

B) All integers less than or equal to 5

C) All nonnegative real numbers

D) All nonnegative integers

Question 2.

[Tg:@abusat_760] If $f(x) = x^2$, and $g(x) = x^2 - 6x + 14$, which of the following best describes the graph of $g(x)$ relative to $f(x)$?

A) Raised 5 units and shifted 3 units to the left

B) Raised 5 units and shifted 3 units to the right

C) Same vertical position shifted 3 units to the left

D) Raised 14 units and shifted 6 units to the right

Question 3.

[Tg:@abusat_760] An object is fired upward at an initial velocity (v_0) of 240 feet per second. The height, $h(t)$, of the object as a function of time is $h(t) = v_0 t - 16t^2$. How long will it take the object to hit the ground after takeoff?

A) 16 seconds

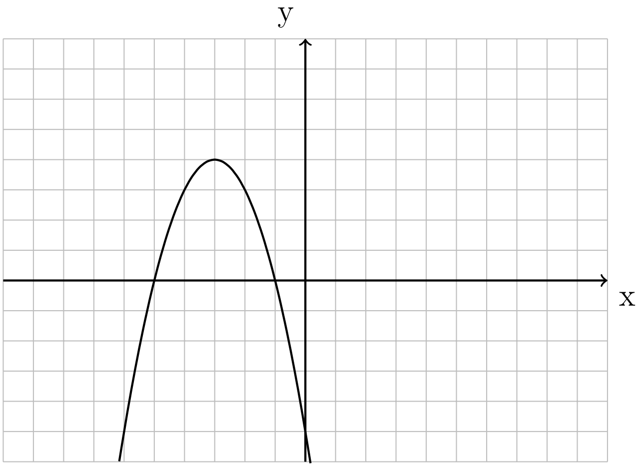
B) 15 seconds

C) 7.5 seconds

D) 4 seconds

Question 4.

[Tg:@abusat_760] What is the equation for the graph below?



A) $y = (x+3)^2 + 4$

B) $y = -x^2 - 6x - 5$

C) $y = (x-3)^2 + 4$

D) $y = -x^2 - 6x - 9$

Question 5.

[Tg:@abusat_760] The quadratic $x^2 - x = 2$ has two solutions. What is the larger of the two solutions?

Question 6.

[Tg:@abusat_760] How is the graph of $g(x) = -x^2$ related to the graph of $f(x) = x^2$?

A) Reflected vertically

B) Reflected horizontally

C) Shifted upwards

D) Shifted to the left

Question 7.

[Tg:@abusat_760] If x is not equal to zero, for what value(s) of x is $x + 1 = \frac{20}{x}$?

A) 4 only

B) -5 only

C) -4 or 5

D) -5 or 4

Question 8.

[Tg:@abusat_760] If $f(x) = ax^2 + bx + c$ is concave downward, which of the following must be true?

A) $a < 0$

B) $a > 0$

C) $b^2 - 4ac > 0$

D) $b^2 - 4ac < 0$

Question 9.

[Tg:@abusat_760] Which of the following describes the graph of the function $f(x) = (-x + 3)(x - 5)$?

A) Opens up with x-intercepts at $(-5, 0)$ and $(3, 0)$

B) Opens up with x-intercepts at $(5, 0)$ and $(3, 0)$

C) Opens down with x-intercepts at $(-5, 0)$ and $(-3, 0)$

D) Opens down with x-intercepts at $(5, 0)$ and $(3, 0)$

Question 10.

[Tg:@abusat_760] If the range of $f(x) = x^2 + 4$ is all real numbers from 13 to 29, what positive numbers lie in the domain of $f(x)$?

A) $3 \leq x \leq 5$

B) $5 \leq x \leq 21$

C) $9 \leq x \leq 25$

D) $13 \leq x \leq 29$

Question 11.

[Tg:@abusat_760] If $x^2 + 7x - 33 = 11$, and $x > 0$, what is the value of $x + 11$?

Question 12.

[Tg:@abusat_760] If A, B, and C are constants such that for all values of x, $x^2 - x - 2 = (Ax + B)(x - 2) + C(x + 3)$, what is the value of A?

A) 1

B) 2

C) 3

D) 4

Question 13.

[Tg:@abusat_760] If the graph of $y = x^2 + mx + n$ passes through the points $(1, 12)$ and $(3, 28)$, what is the value of the product mn?

Question 14.

[Tg:@abusat_760] $y = mx^2 - a$. In the equation above, a is a positive constant and the graph of the equation in the xy-plane is a parabola. Which of the following is an equivalent form of the equation?

A) $y = (mx + a)(mx - a)$

B) $y = (\sqrt{m}x + \sqrt{a})(\sqrt{m}x - \sqrt{a})$

C) $y = (mx + \frac{a}{2})(mx - \frac{a}{2})$

D) $y = (mx - a)^2$

Question 15.

[Tg:@abusat_760] In the xy-plane, the point $(-3, 4)$ lies on the graph of the function f. If $f(x) = k - x^2$, where k is a constant, what is the value of k?

Question 16.

[Tg:@abusat_760] What is the sum of the solutions to $(x - 6)(x + 0.6) = 0$?

A) -6.4

B) -5.6

C) 5.3

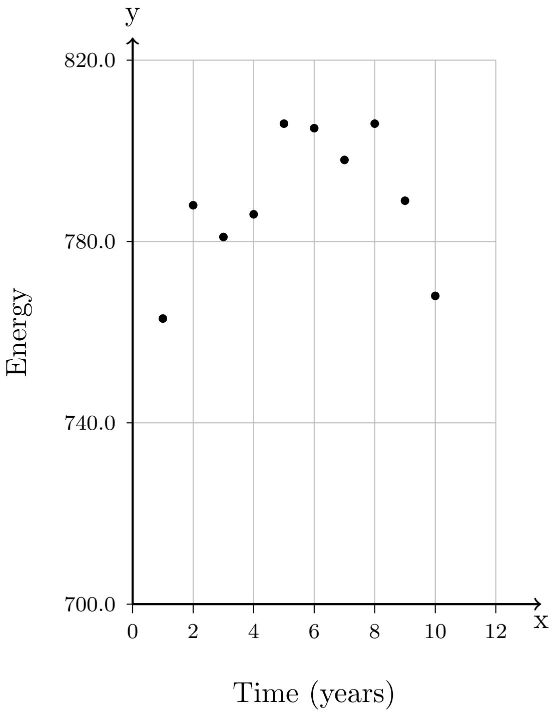
D) 5.4

Question 17.

[Tg:@abusat_760] In the xy-plane, the graph of $y = 3x^2 - 14x$ intersects the graph of $y - x = 0$ at the points $(0, 0)$ and (m, m). What is the value of m?

Question 18.

[Tg:@abusat_760] The scatterplot below shows the amount of electric energy generated, in millions of megawatt-hours, by nuclear sources over a 10-year period.



Of the following equations, which best models the data in the scatterplot?

A) $y = 1.674x^2 + 19.76x - 745.73$

B) $y = -1.674x^2 - 19.76x - 745.73$

C) $y = 1.674x^2 + 19.76x + 745.73$

D) $y = -1.674x^2 + 19.76x + 745.73$

Question 19.

[Tg:@abusat_760] $36 = \frac{3}{2}x^2 - \frac{15}{2}x$. What is the sum of the solutions to the equation above?

A) 2

B) 5

C) 10

D) 23

Question 20.

[Tg:@abusat_760] $y = 7x^2 - 28x + 21$. The graph of the equation above is a parabola in the xy-plane. In which of the following equivalent forms of the equation do the x-intercepts of the parabola appear as constants or coefficients?

A) $y = 7(x^2 - 4x) + 21$

B) $y = 7x(x - 4) + 21$

C) $y = 7(x - 2)^2 - 7$

D) $y = 7(x - 1)(x - 3)$

Question 21.

$$\frac{x^2 - 3x + a}{x - 1}$$

[Tg:@abusat_760] In the expression above, $x \neq 1$ and a is a positive constant. For what value of a is the expression equivalent to $x - 2$?

Question 22.

[Tg:@abusat_760] $a^2 - 8a + 16 = 9$. Based on the equation above, which of the following could be the value of $a - 4$?

A) 8

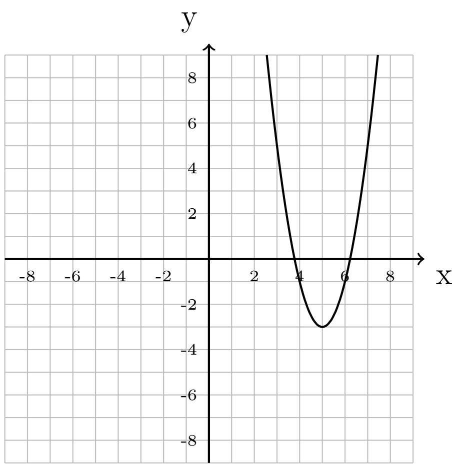
B) 4

C) -3

D) -2

Question 23.

[Tg:@abusat_760] The function f is defined by $f(x) = 2x^2 - 8x + 5$. The graph of $y = f(x - h)$ is shown in the xy-plane.



What is the value of h?

Question 24.

[Tg:@abusat_760] If $t^2 + 6t = 27$ and $t > 0$, what is the value of t?

A) 9

B) 8

C) 3

D) 2

Question 25.

$$(x - k)(x - 2) > 0$$

[Tg:@abusat_760] In the inequality above, k is a constant. If the solution set for the inequality consists of the values of x such that $x < 2$ or $x > 4$, what is the value of k?

A) -4

B) 0

C) 2

D) 4

Question 26.

[Tg:@abusat_760] $m^2 - 144 = mn$. In the equation, if $n > 0$ and both m and n are integers, what is one possible value of n?

Question 27.

$$y = 0$$

$$y = 2x^2 + 12x + c$$

[Tg:@abusat_760] In the system of equations above, c is a positive constant. For which value of c does the system have exactly one real solution?

Question 28.

[Tg:@abusat_760] If $f(x) = x^2 - 4x + 10$ and c is a positive integer less than 5, what is one possible value of $f(c)$?

Question 29.

[**Tg:@abusat_760**] The graphs in the xy-plane of the following quadratic equations each have x-intercepts of -2 and 4. The graph of which equation has its vertex farthest from the x-axis?

A) $y = -7(x + 2)(x - 4)$

B) $y = \frac{1}{10}(x + 2)(x - 4)$

C) $y = -\frac{1}{2}(x + 2)(x - 4)$

D) $y = 5(x + 2)(x - 4)$

Question 30.

[**Tg:@abusat_760**] In the xy-plane, the graph of $y = (x - 6)^2 + 3$ is the image of the graph of $y = (x + 5)^2 + 3$ after a translation of how many units to the right?

Question 31.

[**Tg:@abusat_760**] Which of the following is an example of a function whose graph in the xy-plane has no x-intercepts?

A) A linear function whose rate of change is zero

B) A cubic polynomial with at least one real zero

C) A quadratic function with no real zeros

D) A quadratic function with one real zero

Question 32.

[**Tg:@abusat_760**] What are the solutions of the quadratic equation

$$3x^2 - 6x - 9 = 0?$$

A) $x = -1$ and $x = -3$

B) $x = -1$ and $x = 3$

C) $x = 1$ and $x = -3$

D) $x = 1$ and $x = 3$

Question 33.

[Tg:@abusat_760] $px^2 - t = (4x+3)(4x-3)$. In the equation above, p and t are constants. Which of the following could be the value of p/t?

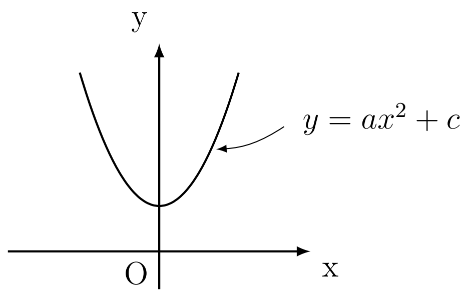
A) 16/3

B) 16/9

C) 4/9

D) 4/3

Question 34.



[Tg:@abusat_760] The vertex of the parabola in the xy-plane above is $(0, c)$. Which of the following is true about the parabola with the equation $y = -[a(x+h)^2 - c]$?

A) The vertex is (h, c) and the graph opens upward.

B) The vertex is (h, c) and the graph opens downward.

C) The vertex is $(-h, -c)$ and the graph opens downward.

D) The vertex is $(-h, c)$ and the graph opens downward.

Question 35.

[Tg:@abusat_760] $6x^2 - 12x = 2t$. In the equation above, t is a constant. If the equation has no real solutions, which of the following could be the value of t?

A) -1

B) -2

C) -3

D) -4

Question 36.

[**Tg:@abusat_760**] $(x-2)^2 + y^2 = 41$. One equation in a system of equations is shown above. The graph of the second equation (not shown) is a vertical line in the xy-plane that crosses the x-axis at $(3, 0)$. For both y-coordinates of the solutions to the system of equations, what is the value of y^2?

A) 58

B) 42

C) 40

D) 7

Question 37.

[**Tg:@abusat_760**] Which of the following equations has exactly one real solution?

A) $x^2 - 6x = -9$

B) $x^2 - 5x = 25$

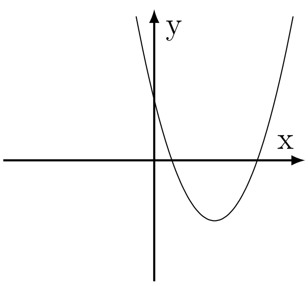
C) $x^2 = 10x + 25$

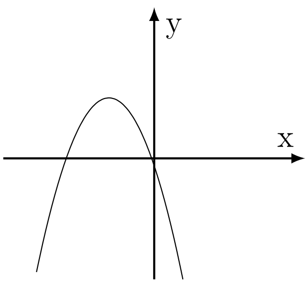
D) $x^2 = 2x + 1$

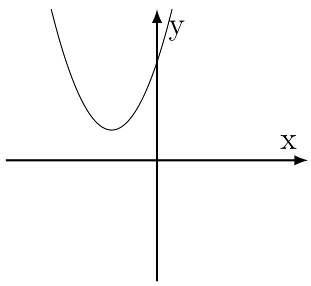
Question 38.

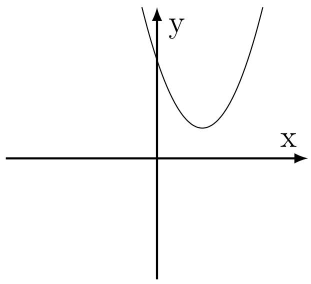
$$f(x) = h((x+h)^2 + 1/2)$$

[**Tg:@abusat_760**] In the function f defined above, $h > 1$. Which of the following best represents the graph of f in the xy-plane?

A)


B)


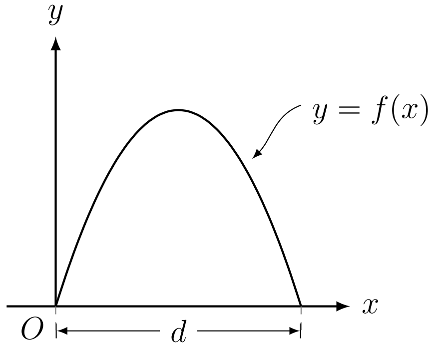
C)


D)


Question 39.

[Tg:@abusat_760] $y = x^2 - 6x + k$. In the equation, k is a constant. If the equation represents a parabola in the xy-plane that is tangent to the x-axis, what is the value of k?

Question 40.



[Tg:@abusat_760] An architect designs an archway by resting the two bases of an arch on the floor. The shape of the arch can be modeled by the graph of the function f in the xy-plane above. If f is defined by $f(x) = -2x^2 + 10x$, where x is in feet, what is the value of the distance d, in feet, between the two bases of the arch?

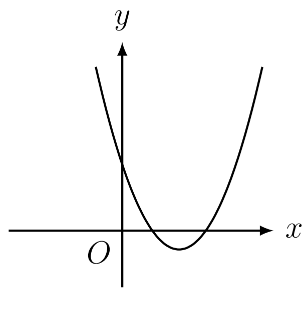
A) 2

B) 5

C) 10

D) 20

Question 41.



[Tg:@abusat_760] Which of the following could be an equation of the parabola in the xy-plane above?

A) $y = x(x + 5)$

B) $y = 2(x + 1.2)(x - 4.1)$

C) $y = 2(x - 1)(x + 4.1)$

D) $y = 2(x - 1.2)(x - 3.5)$

Question 42.

[Tg:@abusat_760] The function f is defined by $f(x) = (x+2)^2$. If $f(x+a) = x^2-8x+16$, where a is a constant. What is the value of a?

A) 8

B) 6

C) -6

D) -8

Question 43.

[Tg:@abusat_760] If x is positive and $x + 2(x - 1) + 3(x - 2) = x^2 - 15$, what is the value of x?

Question 44.

[Tg:@abusat_760] In the function f defined by $f(x) = 2x^2 - kx + 4$, what must $\frac{k}{4}$ represent?

A) The maximum value of f

B) The y-intercept of the graph of f

C) The x-coordinate of the vertex of f

D) The x-intercept of the graph of f

Question 45.

[Tg:@abusat_760] If the graph of the function $f(x)$ is a parabola with vertex $(2, 7)$ and y-intercept $(0, -1)$. Which of the following defines $f(x)$?

a) $f(x) = -2(x - 2)^2 + 7$

b) $f(x) = -(x - 2)^2 + 7$

c) $f(x) = -2(x + 2)^2 + 7$

d) $f(x) = (x - 2)^2 - 7$

Question 46.

[Tg:@abusat_760] $f(x) = a(x-p)(x-q)$. For quadratic function f above, a, p and q are constants. In the xy-plane, the graph of f is a parabola that opens downward, and the coordinates of its vertex are in the second quadrant. Which of the following could be true?

a) $a < 0, p > 0, q > 0$

b) $a > 0, p > 0, q > 0$

c) $a < 0, p > 0, q < 0$

d) $a > 0, p < 0, q < 0$

Question 47.

[Tg:@abusat_760] $3x^2 - 29x + 18 = 0$. What is a solution to the equation above?

Question 48.

[Tg:@abusat_760] What values of x satisfy $x^2 + 5x + 6 = 9x + 3$?

A) 1 and 3

B) 1 and - 3

C) 2 and 3

D) 2 and - 3

Question 49.

$$x^2 - 6x - 10 = 0$$

[Tg:@abusat_760] The quadratic equation above has solutions $x = a + \sqrt{b}$ and $x = a - \sqrt{b}$, where a and b are constants. What is the value of $\frac{b}{a}$?

Question 50.

$$x^2 - 36 = 0$$

[Tg:@abusat_760] If x is a solution of the equation above, which of the following is a possible value of X?

A) 36

B) -4

C) -6

D) 18

Answer Key

1	2	3	4	5	6	7	8	9	10
A	B	B	B	2	A	D	A	D	A

11	12	13	14	15	16	17	18	19	20
15	A	28	B	13	D	5	D	B	D

21	22	23	24	25	26	27	28	29	30
2	C	3	C	D	*	18	6,7,10	A	11

31	32	33	34	35	36	37	38	39	40
C	B	B	D	D	C	A	C	9	B

41	42	43	44	45	46	47	48	49	50
D	C	7	C	A	C	2/3or9	A	19/3	C

* = 7,18,45,70,143